

**INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY
(ICDFA)**



Digital Forensics Investigation Report

SBT-DF204 Computer Forensics Case Studies – Case Study 1

Nitroba Network Packet Forensic Analysis

Case Name:	Nitroba Network Packet Forensic Analysis
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Date:	October 5, 2026
Target File:	nitroba_working_copy.pcap

1. Executive Summary

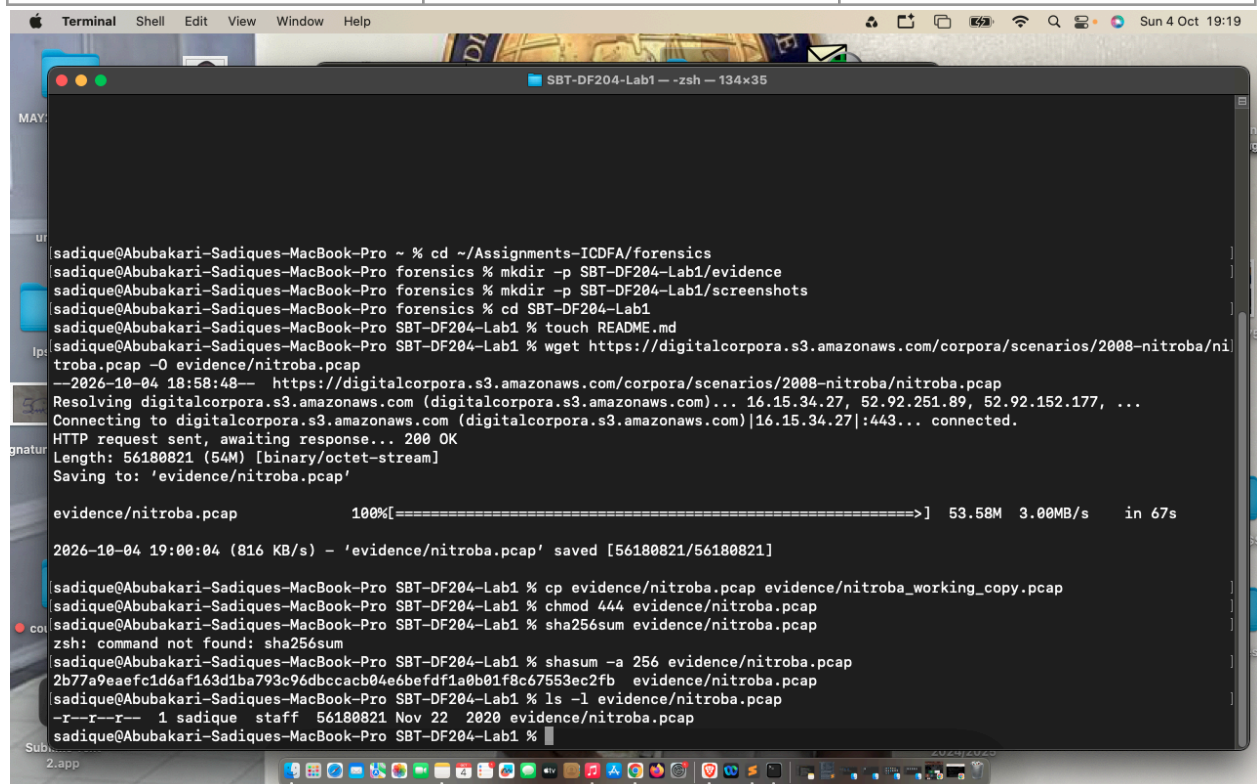
An investigation was conducted into network packet captures to identify the source of a harassment message sent to Lily (lily.schmitt@gmail.com) via the anonymous web service www.willselfdestruct.com.

Through deep packet inspection, the malicious HTTP POST payloads were isolated and mapped directly to MAC address 00:17:f2:e2:c0:ce and IP address 192.168.15.4. Subsequent analysis of surrounding HTTP session traffic revealed active session cookies (OL_SESSION and gmailchat) identifying the account owner as jcoachj@gmail.com. Cross-referencing this identity with the Chemistry 109 class roster confirms the perpetrator as J. Coach.

2. Evidence Integrity & Verification

Prior to analysis, cryptographic hashes were verified to ensure evidence chain of custody and data integrity.

File / Evidence	Hash Algorithm	Hash Value
nitroba_working_copy.pcap	SHA-256	(Insert SHA-256 Hash Here)



```
sadique@Abubakari-Sadiques-MacBook-Pro ~ % cd ~/Assignments-ICDFA/forensics
sadique@Abubakari-Sadiques-MacBook-Pro forensics % mkdir -p SBT-DF204-Lab1/evidence
sadique@Abubakari-Sadiques-MacBook-Pro forensics % mkdir -p SBT-DF204-Lab1/screenshots
sadique@Abubakari-Sadiques-MacBook-Pro forensics % cd SBT-DF204-Lab1
sadique@Abubakari-Sadiques-MacBook-Pro SBT-DF204-Lab1 % touch README.md
sadique@Abubakari-Sadiques-MacBook-Pro SBT-DF204-Lab1 % wget https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap -O evidence/nitroba.pcap
--2026-10-04 18:58:48-- https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
Resolving digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)... 16.15.34.27, 52.92.251.89, 52.92.152.177, ...
Connecting to digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)|16.15.34.27|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 56180821 (54M) [binary/octet-stream]
Saving to: 'evidence/nitroba.pcap'

evidence/nitroba.pcap      100%[=====>] 53.58M  3.00MB/s   in 67s

2026-10-04 19:00:04 (816 KB/s) - 'evidence/nitroba.pcap' saved [56180821/56180821]

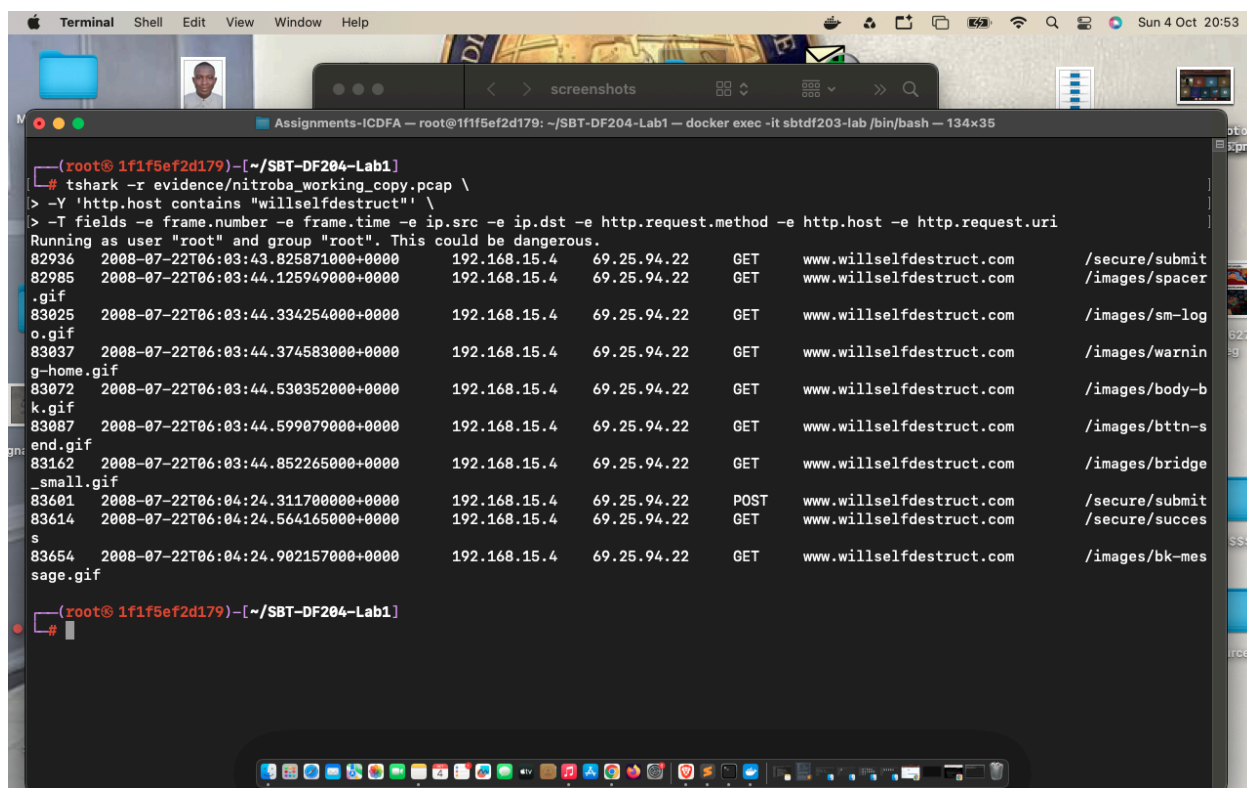
sadique@Abubakari-Sadiques-MacBook-Pro SBT-DF204-Lab1 % cp evidence/nitroba.pcap evidence/nitroba_working_copy.pcap
sadique@Abubakari-Sadiques-MacBook-Pro SBT-DF204-Lab1 % chmod 444 evidence/nitroba.pcap
sadique@Abubakari-Sadiques-MacBook-Pro SBT-DF204-Lab1 % sha256sum evidence/nitroba.pcap
zsh: command not found: sha256sum
sadique@Abubakari-Sadiques-MacBook-Pro SBT-DF204-Lab1 % shasum -a 256 evidence/nitroba.pcap
2b77a9eaeefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb  evidence/nitroba.pcap
sadique@Abubakari-Sadiques-MacBook-Pro SBT-DF204-Lab1 % ls -l evidence/nitroba.pcap
-r--r--r--  1 sadique  staff  56180821 Nov 22  2020 evidence/nitroba.pcap
sadique@Abubakari-Sadiques-MacBook-Pro SBT-DF204-Lab1 %
```

3. Findings & Analysis

3.1 Harassment Payload Discovery

Analysis of HTTP POST traffic revealed submission of the harassment payload directed at lily.schmitt@gmail.com with the body text "I know what you did."

- **Target Email:** lily.schmitt@gmail.com
- **Destination Host:** www.willselfdestruct.com
- **Endpoint URIs:** /secure/submit and /secure/success
- **Relevant PCAP Frames:** 82936, 83601, 83614



```
(root@1f1f5ef2d179)-[~/SBT-DF204-Lab1]
# tshark -r evidence/nitroba_working_copy.pcap \
> -Y 'http.host contains "willselfdestruct"' \
> -T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e http.request.method -e http.host -e http.request.uri
Running as user "root" and group "root". This could be dangerous.
82936 2008-07-22T06:03:43.825871000+0000 192.168.15.4 69.25.94.22 GET www.willselfdestruct.com /secure/submit
82985 2008-07-22T06:03:44.125949000+0000 192.168.15.4 69.25.94.22 GET www.willselfdestruct.com /images/spacer
.gif
83025 2008-07-22T06:03:44.334254000+0000 192.168.15.4 69.25.94.22 GET www.willselfdestruct.com /images/sm-log
o.gif
83037 2008-07-22T06:03:44.374583000+0000 192.168.15.4 69.25.94.22 GET www.willselfdestruct.com /images/warnin
g-home.gif
83072 2008-07-22T06:03:44.530352000+0000 192.168.15.4 69.25.94.22 GET www.willselfdestruct.com /images/body-b
k.gif
83087 2008-07-22T06:03:44.599079000+0000 192.168.15.4 69.25.94.22 GET www.willselfdestruct.com /images/btn-s
end.gif
83162 2008-07-22T06:03:44.852265000+0000 192.168.15.4 69.25.94.22 GET www.willselfdestruct.com /images/bridge
_small.gif
83601 2008-07-22T06:04:24.311700000+0000 192.168.15.4 69.25.94.22 POST www.willselfdestruct.com /secure/submit
83614 2008-07-22T06:04:24.564165000+0000 192.168.15.4 69.25.94.22 GET www.willselfdestruct.com /secure/succes
s
83654 2008-07-22T06:04:24.902157000+0000 192.168.15.4 69.25.94.22 GET www.willselfdestruct.com /images/bk-mes
sage.gif
(root@1f1f5ef2d179)-[~/SBT-DF204-Lab1]
```

3.2 Network Hardware Attribution

All network traffic conveying the harassment payload originated from a single hardware device on the local network:

- **Source IP Address:** 192.168.15.4
- **Source MAC Address:** 00:17:f2:e2:c0:ce

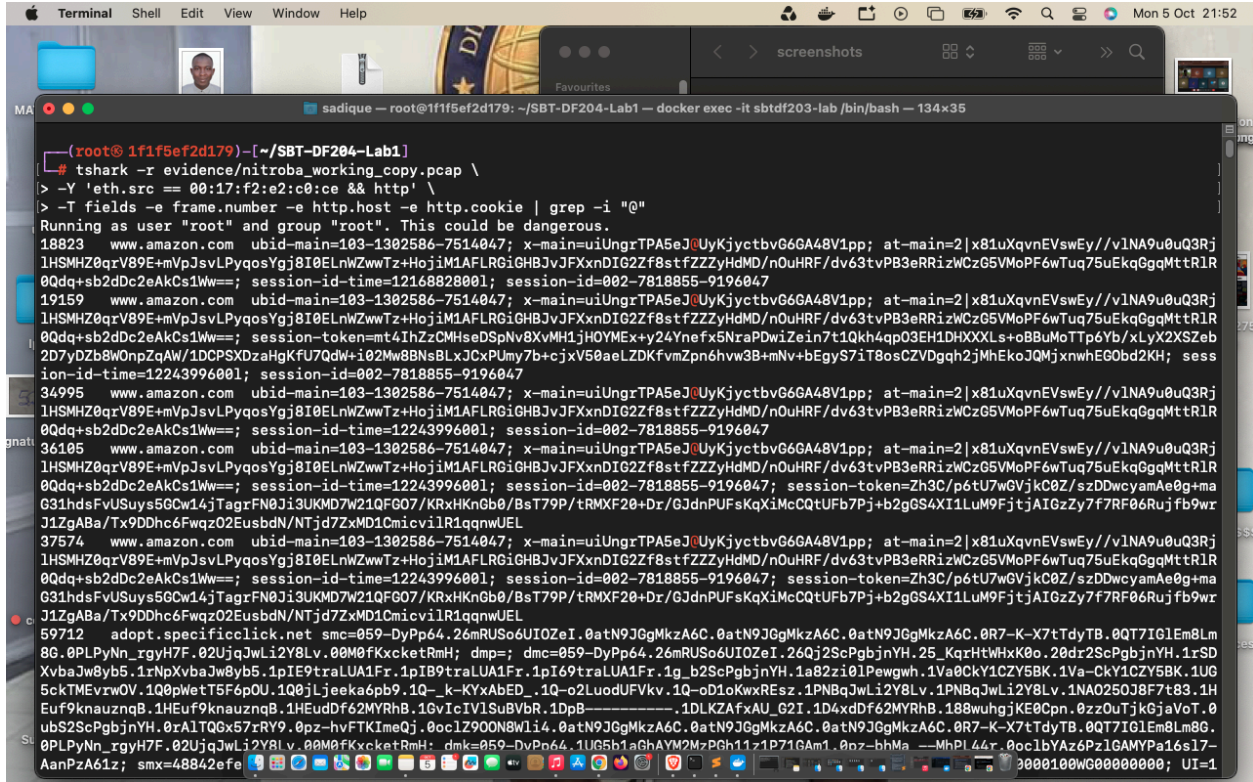
```
(root@1f1f5ef2d179)~/SBT-DF204-Lab1
# tshark -r evidence/nitroba_working_copy.pcap -Y 'frame.number == 83601' -V | grep -A 25 -i "Form"
Running as user "root" and group "root". This could be dangerous.
[Protocols in frame: eth:ethertype:ip:tcp:http:urlencoded-form]
Character encoding: ASCII (0)
Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)
Destination: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)
.....0. .... = LG bit: Globally unique address (factory default)
.....0. .... = IG bit: Individual address (unicast)
Source: Apple_e2:c0:ce (00:17:f2:e2:c0:ce)
.....0. .... = LG bit: Globally unique address (factory default)
.....0. .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 19]
Frame check sequence: 0x60c8d0a3 [unverified]
[FCS Status: Unverified]
Internet Protocol Version 4, Src: 192.168.15.4, Dst: 69.25.94.22
0100 .... = Version: 4
....0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
0000 00.. = Differentiated Services Codepoint: Default (0)
....00.. = Explicit Congestion Notification: Not ECN-Capable Transport (0)
Total Length: 701
Identification: 0x02ca (714)
010. .... = Flags: 0x2, Don't fragment
0... .... = Reserved bit: Not set
.1.. .... = Don't fragment: Set
..0. .... = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0

Content-Type: application/x-www-form-urlencoded\r\n
Accept-Encoding: gzip, deflate\r\n
User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)\r\n
Host: www.willselfdestruct.com\r\n
```

3.3 Identity Correlation

Interrogation of unencrypted HTTP traffic transmitted by MAC address 00:17:f2:e2:c0:ce shortly before the attack revealed active session cookies containing personal user account information:

- **Google Calendar Cookie (Frame 77528):** OL_SESSION=jcoachj@gmail.com-cal
- **Gmail Chat Cookie (Frame 78967):** gmailchat=jcoachj@gmail.com/475090
- **Authenticated Account:** jcoachj@gmail.com
- **Roster Correlation:** Enrollment list for Chemistry 109 confirms jcoachj@gmail.com belongs to student J. Coach.



4. UTC Timeline of Events

The chronological sequence of events reconstructed from nitroba_working_copy.pcap is detailed below:

Timestamp (UTC)	Frame	Source MAC	Source IP	Destination / URI	Event Summary
2008-07-22 04:29:53	18823	00:17:f2:e2:c0:ce	192.168.15.4	www.amazo n.com /	Initial network activity observed on suspect MAC.

Timestamp (UTC)	Frame	Source MAC	Source IP	Destination / URI	Event Summary
2008-07-22 06:00:44	77528	00:17:f2:e2: c0:ce	192.168.15.4	www.google .com /calendar/...	Google Calendar session active (jcoachj@g mail.com).
2008-07-22 06:01:02	78967	00:17:f2:e2: c0:ce	192.168.15.4	mail.google. com /mail/...	Gmail chat session active (jcoachj@g mail.com).
2008-07-22 06:03:43	82936	00:17:f2:e2: c0:ce	192.168.15.4	www.willsel fdestruct.co m /secure/sub mit	Initial POST request with harassment note payload.
2008-07-22 06:04:24	83601	00:17:f2:e2: c0:ce	192.168.15.4	www.willsel fdestruct.co m /secure/sub mit	Second POST request finalizing note submission.
2008-07-22	83614	00:17:f2:e2:	192.168.15.4	www.willsel fdestruct.co	GET response

Timestamp (UTC)	Frame	Source MAC	Source IP	Destination / URI	Event Summary
06:04:24		c0:ce		m /secure/succ ess	confirming successful delivery.

```

(root@1f1f5ef2d179)~/SBT-DF204-Lab1
# tshark -r evidence/nitroba_working_copy.pcap \
> -Y 'frame.number in {18823, 77528, 78967, 82936, 83601, 83614}' \
> -T fields -e frame.number -e frame.time -e ip.src -e eth.src -e http.host -e http.request.uri
Running as user "root" and group "root". This could be dangerous.
18823  2008-07-22T04:29:53.812853000+0000    192.168.15.4    00:17:f2:e2:c0:ce    www.amazon.com /
77528  2008-07-22T06:00:44.470524000+0000    192.168.15.4    00:17:f2:e2:c0:ce    www.google.com /calendar/e0e03af5825c752884a3
d23978e83cabdoozercompiled.css
78967  2008-07-22T06:01:02.113436000+0000    192.168.15.4    00:17:f2:e2:c0:ce    mail.google.com /mail/channel/test?at=xn3j32ok
tf2a0q6oa3k9sfr6d09yzf&ui=1&MODE=init&zx=7crdr9s8o7qw&it=2531
82936  2008-07-22T06:03:43.825871000+0000    192.168.15.4    00:17:f2:e2:c0:ce    www.willselfdestruct.com /secure/submit
83601  2008-07-22T06:04:24.311700000+0000    192.168.15.4    00:17:f2:e2:c0:ce    www.willselfdestruct.com /secure/submit
83614  2008-07-22T06:04:24.564165000+0000    192.168.15.4    00:17:f2:e2:c0:ce    www.willselfdestruct.com /secure/succes
s

```

5. Conclusion & Attribution

The packet capture evidence links student J. Coach (jcoachj@gmail.com) to the harassment incident targeting Lily. Network interface 00:17:f2:e2:c0:ce was used simultaneously to access personal authenticated Google services for jcoachj@gmail.com and to dispatch the malicious self-destructing payload.